Ochsner Health

Critical Care & Emergency Drug Guide

Includes COVID-19 Emergency Shortage Medications

Version 5- Created 2022
Approved for use by: System Pharmacy (P&T) 8/22, OMC-NO NPD 9/22, SNPD 8/22
For changes or edits please contact Shannon West (shwest@ochsner.org) for Lee Ann Dooley (ldooley@ochsner.org)

Antihypertensives Page 1

Name	Concentrations	Dose	Onset of Action	Duration of Action	Comments
clevidipine (Cleviprex)	25 mg/50 mL 50 mg/100 mL (0.5 mg/mL)	**Initial:** 1-2 mg/hr **Titrate up:** Double the dose Q 2 minutes **Max:** 16 mg/hr (can use doses up to 32 mg/hr short term) **Titrate down:** 1-2mg/hr Q 5 minutes	2-4 minutes	5-15 minutes	• Look alike medication with propofol (Diprivan) • Max dose over 24 hrs. is 1,000 mL due to lipid content • Use or discard within 12 hours of puncturing stopper • **Change IV tubing q 12 hours** • Check compatibility before administering anything in the same line – **most meds not compatible.** • Contraindicated for patients with soy or egg allergies • Monitor for rebound HTN after stopping the infusion
enalapril (Vasotec)	n/a	1.25-5 mg IVP Q 6 hours	15-30 minutes	6-12 hours	• Response may vary • Avoid in acute MI
hydrALAZINE (Apresoline)	n/a	10-20 mg IVP Q 1-4 hours **Max:** 20 mg per dose	10-20 minutes	1-4 hours	• Preferred agent for eclampsia • Causes headache, tachycardia, flushing
labetalol (Normodyne, Trandate)	500 mg/100 mL (5 mg/mL) **D5W or NS**	10-80 mg IVP Q 10-15 minutes **Usual**: 10-20 mg IVP Q 10 minutes **IV infusion:** 0.5-2 mg/minute **Titrate**: 0.1 mg/min Q 15 minutes **Max**: 3 mg/min	5-10 minutes	3-6 hours	• Long duration – if HR/BP too low can take hours to wear off • May cause bronchospasm • **Contraindicated in severe asthma, bradycardia, heart blocks, decompensated heart failure**
metoprolol (Lopressor)	n/a	2.5-10 mg IVP over 3 minutes Q 5 minutes **Max cumulative dose:** 15 mg	5 minutes	Varies Few minutes – 6 hours	• Will decrease BP & HR (ß1 selective) • Some risk of heart block • 5 mg IV Q 6H = 25 mg PO Q 12H

Antihypertensives Page 2

Name	Concentrations	Dose	Onset of Action	Duration of Action	• Comments
niCARdipine (Cardene)	**Standard** 40 mg/200 mL 50 mg/250 mL (0.2 mg/mL) **High** 125 mg/250 mL (0.5 mg/mL) **NS**	**Initial:** 2.5-5 mg/hr IV infusion **Titrate**: 2.5 mg/hr Q 5-15 minutes **Max**: 15 mg/hr	5-10 minutes	15-30 minutes	• Can cause a reflex tachycardia • Caution with cardiac ischemia • Pure arterial vasodilator • Central line preferred
nitroGLYcerin (Tridil) **Screen patient for use of erectile dysfunction or pulmonary HTN meds**	50 mg/250 mL (200 mcg/mL) 100 mg/250mL (400mcg/mL) **D5W** **Must go in Low-Sorb (blue) tubing**	**Initial IV Infusion**: 5-20 mcg/minute **Titrate:** 5-20 mcg/minute Q 3-5 minutes **Max:** 400 mcg/minute	2-5 minutes	5-10 minutes	• Preferred agent for coronary ischemia and flash pulmonary edema • Can cause headache • Tolerance with prolonged use • Pure venous vasodilator • **Caution with high ICP**
nitroPRUSSide (Nipride)	**High:** 50 mg/100 mL (0.5 mg/mL) **D5W only** **Protect from light** **Must go in Low-Sorb (blue) tubing**	**Initial:** 0.3 mcg/kg/minute **Titrate:** 0.2 mcg/kg/min Q3-5 minutes **Max:** 10 mcg/kg/min	immediate	1-2 minutes	• Cyanide toxicity with doses > 3 mcg/kg/minute for prolonged periods • Blood thiocyanate levels daily if dose > 3 mcg/kg/min (1 mcg/kg/min if anuric) • Caution with high ICP • Avoid in renal impairment

Anti-Arrhythmics – Page 1

Name	Concentrations	Dose	Onset of Action	Duration of Action	Comments
adenosine (Adenocard)	6 mg/2 mL 12 mg/4 mL	6 mg **RAPID** IVP If not effective within 1-2 minutes, may give 12 mg **RAPID** IVP May repeat 12 mg IVP if needed	Immediate Interrupts re-entry pathway through AV node	<10 seconds	• **Use 2 syringes** (one med/one NS flush) connected to a T-connector or stopcock using most proximal site • Follow each IVP with RAPID 20 ml NS flush • Not for WPW syndrome patients • EKG monitoring and defibrillator with pads placed required for administration • Central Line Administration: Use half the dose
AMIODArone (Cordarone) **Use 0.22 micron in-line filter during administration**	**Standard** 360 mg/200 mL (1.8 mg/mL) **High** 900 mg/250 mL (3.6 mg/mL-Central Line only) **D5W Premix**	**Loading Dose:** **Stable patient**: 150 mg/100 mL infusion over 10 minutes (can repeat x 1) **Unstable patient**: 300 mg IVP **Infusion:** 1 mg/minute x 6 hours then 0.5 mg/minute x 18 hrs OR until PO conversion	IV: rapid Oral: days	several days	• Central line administration preferred • Short-term – large gauge peripheral okay • After 24-hour infusion, convert to oral **OR** continue at 0.5 mg/minute (total loading dose of 10 g) • IV formulation may lower HR and BP • Many drug interactions • Can cause pulmonary, thyroid, dermatologic, and hepatic toxicity
digoxin (Lanoxin)	n/a	**Load**: 500 mcg IVP followed by 250 mcg IVP Q6 hours x 2 (**total**: 1 mg) **Low weight patient:** 8-12 mcg/kg – give 50% x 1 dose then 25% Q 6 hours x 2 doses	5-60 minutes	1-2 days	• Lower loading dose by 50% in ESRD • Renal function will determine maintenance • Used for heart rate control • Goal level 0.5 – 2 ng/mL (lower goal for HF, higher goal for atrial fibrillation)
dilTIAZem (Cardizem)	100 mg/100 mL 125 mg/125 mL (1 mg/mL) **D5W**	**IVP** 10-20 mg -may repeat x 1 **IV infusion:** Initial: 2.5-5 mg/hr **Titrate:** 2.5 mg/hr Q15 minutes **Max:** 15 mg/hr	2-7 minutes	30 minutes – 2 hours	• Will decrease HR and BP • Should D/C infusion 2-3 hours after first PO dose

Anti-Arrhythmics – Page 2

Name	Concentrations	Dose	Onset of Action	Duration of Action	Comments
esmolol (Brevibloc)	2000 mg/100 mL (20 mg/mL) **NS Premix**	**Initial:** 50 mcg/kg/minute **Titrate:** 25 mcg/kg/minute Q 5 minutes **Max:** 300 mcg/kg/minute	1-2 minutes	10-30 minutes	• Preferred agent in aortic dissection • Not great at BP control, better as heart **rate control** drug • Do not discontinue abruptly • Central line preferred • Transfer to oral agent ASAP – decrease infusion by 50% 30 minutes after first oral agent
LIDOcaine	2000 mg/250 mL (8 mg/mL) **D5W Premix**	**Load:** 1-1.5 mg/kg IVP over 2-3 min (usual 100 mg) may repeat x 1 **Max cumulative bolus:** 3 mg/kg **Infusion:** **Initial:** 1 mg/min **Titrate**: 0.5 mg/min Q 10 minutes **Max:** 4 mg/min	45-90 seconds	1 hour	• May cause seizures (especially rapid administration), respiratory arrest, cardiac toxicity • If arrhythmias occur during infusion, give 0.5 mg/kg bolus to increase plasma concentration of the drug • Convert to oral agent after 24 hours if able • Recommend level if infusion >12-24 hours or if patient has hepatic dysfunction
procainamide (Pronestyl)	2000 mg/250 mL (8 mg/mL) **D5W**	**Load:** 20-50 mg/minute IV infusion until arrhythmia controlled or max dose (1000 mg) reached **Max Bolus Dose**: 1000 mg **Maintenance Infusion**: **Initial:** 1 mg/min **Max:** 4 mg/min	10-30 minutes	6-48 hours	• If rhythm converts, decrease to maintenance infusion • If QRS widens by >50% OR hypotension → decrease rate; may need to stop the drip • Drug of choice in Wolfe-Parkinson White Syndrome (WPW) • Recommend monitoring for patients with renal dysfunction and infusions > 24 hours • Caution on initial dose/titrations in renal dysfunction

Adult Emergency/Critical Care Medication Reference

**** Please Note: Not all medications are available at each institution****

Vasopressors *Central Line Preferred for all*

Name	Concentrations	Dose	Onset of Action	Duration of Action	Comments
DOPamine	**Standard** 400 mg/250 mL (1.6 mg/mL) **High** 800 mg/250 mL (3.2 mg/mL) **D5W**	**Initial:** 2.5-10 mcg/kg/min **Titrate:** 2.5-5 mcg/kg/min Q 5 min **Max:** 20 mcg/kg/minute	5 minutes	10 minutes	• Used for refractory bradycardia • Increases HR, contractility, and cardiac output • **Low/Intermediate Dose (1-9 mcg/kg/min)** • **High doses (≥10 mcg/kg/min)**: Increase BP & SVR • May cause tachycardia &/or arrhythmias
EPINEPHrine	**Standard** 5 mg/250 mL (20 mcg/mL) **High** 10 mg/250 mL (40 mcg/mL) 30 mg/250 mL (120 mcg/mL) **NS**	**Initial:** 0.02-0.2 mcg/kg/minute **Titration:** 0.02 mcg/kg/min Q 5 min **Max**: 2 mcg/kg/minute	immediate	5 minutes	• Increases HR, SVR, & CO • May cause tachycardia, arrhythmias, myocardial ischemia
NORepinephrine (Levophed)	**Standard** 4 mg/250 mL (16 mcg/mL) 8 mg/250 mL (32 mcg/mL) **High:** 32 mg/250 mL (128 mcg/mL) **D5W**	**Initial:** 0.02-0.2 mcg/kg/min **Titration:** 0.02 mcg/kg/minute Q 5 min **Max**: 3 mcg/kg/minute	immediate	2 minutes	• Increases SVR, HR, CO, & contractility • Drug of choice for sepsis: Goal MAP ≥ 65 • Any concentration > 4 mg/250 mL must be infused via central line
PHENYLephrine (NeoSynephrine)	**Standard** 20 mg/250 mL (80 mcg/mL) **High** 100 mg/250 mL (400 mcg/mL) **NS**	**Initial:** 0.5 mcg/kg/minute **Titration:** 0.5 mcg/kg/minute Q 5 min **Max**: 5 mcg/kg/minute	immediate	15 minutes	• Increases SVR without increased HR or contractility changes • Can cause reflex bradycardia **Neo-Bump: (for emergent hypotension)** ➢ Initial dose is 0.05-0.1 mg (50-100 mcg) ➢ May repeat Q 5-15 minutes
vasopressin	**Standard:** 20 units/100 mL (0.2 units/mL) **High:** 100 units/100 mL (1 unit/mL) **D5W**	0.04 units/minute **No titration**	3-5 minutes	10-20 minutes	• Increases SVR • Second line pressor for sepsis • Direct vasoconstrictor • Non-weight-based dosing
angiotension II (Giapreza)	**2.5 ng/250 mL** (10,000 ng/mL) **NS**	**Initial:** 10 **ng**/kg/min **Titration:** 5-15 **ng**/kg/min Q 5 min **Max:** In first 3 hours – 80 **ng**/kg/min After 3 hours – 40 **ng**/kg/min	immediate	<1 minute	• 3rd line pressor for distributive shock in adults after norepinephrine dose ≥ 0.2mcg/kg/min + vasopressin at 0.04 units/min with MAP < 65 • S/E: increased risk for blood clots – VTE prophylaxis required • **Restricted use in certain clinical areas/specialties**

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Inotropes

Name	Concentrations	Dose	Onset of Action	Duration of Action	Comments
DOBUTamine (Dobutrex)	1000 mg/ 250 mL (4 mg/mL) **D5W Premix**	2-10 mcg/kg/minute IV infusion **Titration:** 2.5 mcg/kg/minute Q 15 minutes **Max:** 20 mcg/kg/minute	2 minutes	10 minutes	• Increases CO, contractility, and HR • May decrease SVR (afterload) • May cause hypotension • Can cause new or worsening arrhythmias, angina, tachycardia • Low, non-titrating doses used for cardiac decompensation in heart failure
isoproterenol (Isuprel)	1 mg/250 mL (0.004 mg/mL) **D5W**	2 mcg/minute IV infusion **Titration:** 0.5-1 mcg/min Q 5 minutes **Max**: 10 mcg/min IV infusion	Immediate	10 -15 minutes	• 2nd line agent for sustained bradycardia unresponsive to dopamine • Increases HR and contractility • Decreases SVR • $$$
milrinone (Primacor)	**Standard** 20 mg/100 mL (0.2 mg/mL) **High** 40 mg/100 mL (0.4 mg/mL) **D5W**	0.25-0.75 mcg/kg/minute IV infusion	30-45 minutes	4 hours	• Increases CO without increasing HR • Decreases SVR • May cause hypotension • Long duration: if BP lowers; hrs to wear off • Accumulates in renal impairment • Used for inotropic support in acute decompensated heart failure • Can cause SVT, VT and dysrhythmias

Analgesics

Different Dosing Acceptable in Palliative Care Patients

Name	Concentrations	Intermittent	Infusion	Onset of Action	Duration of Action	Comments
fenta**NYL**	2500 mcg/250 mL (10 mcg/mL) **D5W**	25-100 mcg IVP Over 3-5 minutes Q 0.5-1hour PRN	**Initial:** 25-100 mcg/hour **Titrate**: 12.5 mcg/hr Q 30 minutes **Max**: 500 mcg/hr	immediate	1-2 hours	• Very little hemodynamic effect • **Drug of Choice** if patient unstable or intubated, especially burns or trauma • Side effects: respiratory depression & constipation • Rapid bolus may cause apnea • **Reversal agent: Naloxone (Narcan)**
HYDROmorphone (Dilaudid)	20 mg/100 mL (0.2 mg/mL) **D5W**	0.5-2 mg IVP Over 2-3 minutes Q2 hours PRN Administer slow IVP to avoid hypotension and respiratory depression	**Initial**: 0.25-0.5 mg/hr **Titrate**: 0.25 mg/hr Q 30 minutes **Max**: 2 mg/hr	5 minutes	3-5 hours	• Side effects: respiratory depression, constipation, and hypotension • Reduce dose by 25-50% in renal patients • Infusion primarily used in palliative care • Preferred over morphine for patients with renal impairment • **Reversal agent: Naloxone (Narcan)**
morphine	100 mg/100 mL (1 mg/mL) **D5W**	0.5-10 mg IVP Dilute in ≥5 mL NS and give over 2-5 minutes Q 1-6 hours PRN	**Initial:** 0.5-2 mg/hr **Titrate**: 0.5 mg/hr Q 30 minutes **Max**: 10 mg/hour	5-10 minutes	3-5 hours	• May cause hypotension and itching due to histamine release • Side effects: respiratory depression & constipation • Reduce dose in renal impairment (active metabolite accumulation) • Infusion primarily used in palliative care • **Reversal agent: Naloxone (Narcan)**
REMIfentanil (Ultiva)	5000 mcg/ 100 mL (50 mcg/mL)	n/a	**Initial:** 0.05 mcg/kg/min **Titrate:** 0.025 mcg/kg/min Q 5 minutes **Max:** 0.2 mcg/kg/min	1-3 minutes	Short 3-10 minute half life	• Common S/E: hypotension, arrhythmias, N/V, muscle rigidity, headache • Renal excretion – monitor effects in patients with decreased renal function or on dialysis

Sedatives

****Administration of sedation should follow your hospital policy on Procedural Sedation/Analgesia for Non-Anesthesiology Providers****

Name	Concentrations	Intermittent	Infusion	Pharmacokinetics	Comments
dexmedeTOMline (Precedex)	200 mcg/50 mL 400 mcg/100 mL 1000 mcg/250 mL (4 mcg/mL) **D5W or NS**	n/a **Do not bolus**	**Initial:** 0.2-0.5 mcg/kg/hr **Titration:** 0.1 mcg/kg/hr Q 30 minutes **Max:** 1.4 mcg/kg/hr	**Onset**: 15-30 minutes **Duration**: 15-120 min dose dependent	• May cause hypotension or bradycardia • Does not cause respiratory depression – okay to use in non-intubated patients
etomidate (Amidate)	n/a	0.15-0.3 mg/kg **IVP by RN for emergent RSI** **One-time dose only**	n/a	**Onset**: 0.5-1 minutes **Duration**: 4-10 min	• Drug of Choice in RSI (Rapid Sequence Intubation) due to less hemodynamic effects than benzodiazepines
ketamine (Ketalar) **RN's may administer anesthetic agents to intubated patients in critical care settings who are ventilated with a control rate of breathing (not pressure support only)**	**Sedation Standard** 500 mg/100 mL (5 mg/mL) **High** 500 mg/50 mL (10 mg/mL) **NS**	**IV:** **Sub-anesthetic/adjunctive:** 0.25-0.5 mg/kg IVP **Sedation:** 1-2 mg/kg IV Push **Intranasal:** **sub-anesthetic/adjunctive:** 0.5-1 mg/kg	**Initial:** 2.5 mcg/kg/min **Titration:** 2.5 mcg/kg/min Q 15 minutes **Max:** 20 mcg/kg/min	**IV:** Onset: 30 seconds Duration: 5-10 minutes **Intranasal:** Onset: 10 minutes Duration: 60 minutes	• Adjunctive dose used for conscious sedation with propofol • **Patient MUST be intubated or have permanent protected airway (trach) with mechanical ventilation on a controlled rate of breathing (not pressure support only)** • Sub-anesthetic doses for pain control can reduce opioid requirement • S/E: tachyarrhythmias, hallucinations and agitation, increased oral secretions, lacrimation (crying), dilated pupils, increased muscle tone (rigidity) • Use for pain management or in patients with an unsecured airway requires initiation, titration, and discontinuation by a provider and is not in the RN's scope of practice in Louisiana.

Sedatives

Administration of sedation should follow your hospital policy on Procedural Sedation/Analgesia for Non-Anesthesiology Providers

Name	Concentrations	Intermittent	Infusion	Pharmacokinetics	Comments
LORazepam (Ativan)	50 mg/50 mL 100 mg/100 mL (1 mg/mL) **D5W**	1-4 mg IVP diluted in equal volume of diluent and given over 2-5 minutes Q 1-6 hours PRN May repeat dosing Q 5-10 min for seizure control	**Initial:** 0.5-2 mg/hour **Titration:** 0.5 mg/hr Q 30 minutes **Max:** 5 mg/hr	**Onset**: 5-20 minutes **Duration**: 4-6 hours	• Solvent can induce **acidosis (propylene glycol)** • May see higher dosing in difficult to control seizures
midazolam (Versed)	**Standard** 50 mg/50 mL 100 mg/100 mL (1 mg/mL) **High** 250 mg/50 mL 500 mg/100 mL (5 mg/mL) **D5W**	1-6 mg IVP Over 2 minutes **OR** 2.5-5 mg IM	**Initial:** 0.5-2 mg/hour **Titration:** 0.5 mg/hr Q 30 minutes **Max:** 5 mg/hr	**Onset**: 2-5 minutes **Duration**: 2-4 hours	• Active metabolite can cause prolonged sedation • Patients should be continuously monitored for hypoventilation, airway obstruction, or apnea • May see higher dosing in status epilepticus
propofol (Diprivan)	1000 mg/100 mL (10 mg/mL) **lipid emulsion** **Bolus or IVP by RN if mechanically ventilated with a controlled rate of breathing (not pressure support only)**	1-2 mg/kg IVP titrated to effect by MD	**Initial:** 5-10 mcg/kg/min **Titrate:** 5 mcg/kg/minute Q 5 minutes **Max**: 50 mcg/kg/min	**Onset**: 1-2 minutes **Duration**: 8 minutes	• Can cause hypotension • High dose can cause increased triglycerides, pancreatitis, propofol infusion syndrome • **Change IV tubing q 12h** • May see higher dosing if used for deep sedation or seizure control – dose should be ordered as **non-titrating**

Neuromuscular Blocking Agents *Must Protect Airway*

Do not administer without sedation!

Do not allow sedation vacations on paralyzed patients

Name	Concentrations	Intermittent	Infusion	Pharmacokinetics	Comments
CISatracurium (Nimbex)	200 mg/100 mL 400 mg/200 mL (2 mg/mL) **D5W**	0.15-0.2 mg/kg from bag **Bolus or IVP by RN if mechanically ventilated with a controlled rate of breathing (not pressure support only)**	**Initial:** 1-3 mcg/kg/minute **Titrate**: 0.5 mcg/kg/minute Q 15 minutes **Max**: 10 mcg/kg/minute	**Onset**: 5-10 minutes **Duration**: 35-45 minutes	• Drug of choice in organ dysfunction • Must get baseline peripheral nerve stimulation (train of four (TOF)) before initiating bolus or infusion if TOF monitoring appropriate for the patient while paralyzed
ROCuronium (Zemuron)	n/a	0.6-1.2 mg/kg IVP **IVP by RN for emergent RSI or if mechanically ventilated with a controlled rate of breathing (not pressure support only)**		**Onset**: 1-2 minutes **Duration**: 30 minutes	• Second line agent for RSI • Long duration of action, additional sedation may be required
succinylcholine	n/a	1-1.5 mg/kg IVP **IVP for emergent RSI by RN**	n/a	**Onset**: 1-2 minutes **Duration**: 10 minutes	• **Drug of Choice** in RSI • **Contraindicated** in hyperkalemia • **Avoid in patients with burns/crush trauma & patients found down for unknown duration** • Can induce **malignant hyperthermia**
VECuronium (Norcuron)	100 mg/100 mL (1 mg/mL) **D5W**	0.08-0.1 mg/kg **Bolus or IVP by RN for emergent RSI or if mechanically ventilated with a controlled rate of breathing (not pressure support only)**	**Initial:** 0.5-1 mcg/kg/min **Titrate**: 0.5 mcg/kg/min Q 30 min **Max:** 2 mcg/kg/min	**Onset**: 5-10 minutes **Duration**: 30-45 minutes	• 50% parent drug and most active metabolite cleared by kidney • Must get baseline peripheral nerve stimulation (train of four) before initiating bolus or infusion if TOF monitoring appropriate for the patient while paralyzed

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IV Anticoagulants

Name	Concentrations	Dose	Infusion	Pharmacokinetics	Comments
argatroban	50 mg/50 mL 125 mg/125 mL (1 mg/mL) **NS**		**Initial:** 0.5 mcg/kg/min (organ dysfunction, post op critically ill) 1-2 mcg/kg/min for stable patients **Titrate:** based on nomogram in orderset **Max:** 10 mcg/kg/min	**Onset:** Immediate **Duration:** 1-3 hours	• Commonly used for HIT+ patients (Heparin Induced Thrombocytopenia) • **Use caution with hepatically impaired patients** • Monitor for bleeding • Max dosing weight 150 kg
bivalirudin (Angiomax)	250 mg/50 mL **D5W**	Bolus 0.75 mg/kg immediately before PCI Check ACT in 5 minutes May give additional 0.3 mg/kg to achieve desired ACT	1.75 mg/kg/hr during procedure and up to 4 hours post then decrease to 0.2 mg/kg/hr for up to 20 hours	**Onset:** immediate **Duration:** 1 hour	• Decrease dose for ESRD or dialysis patient • Monitor for bleeding
heparin	25,000 units/250 mL **D5W Premix**	**Refer to nomogram Must have separate order for bolus – Administer bolus doses from bag via IV pump**	**Weight-based Initial: 12-18 units/kg/hour** **Titrate:** based on nomogram	**Onset:** Immediate **Duration:** 1-2 hours	• Dosage regulated by frequent labs (aPTT) Q6 hours during infusion • Dosing changes take 4-6 hours to affect labs • Dosing based on adjusted body weight • **Ensure correct nomogram is used based on indication (High, Low, Minimum)** • Monitor for bleeding • **Reversal agent: Protamine Sulfate**

Hyperosmolar Therapy for ICP Management

Name	Concentrations	Dose	Onset/Duration	Comments
2% and 3% sodium chloride (2% only at Jeff Hwy)	3% sodium chloride commercially available 2% and buffered solutions must be mixed in pharmacy	Infusion titrated to goal sodium range 3% may be bolused 200-350 mL over 5-20 minutes	Onset: Rapid **Duration:** 1.5 – 4 hours	• **Central Line Preferred for 3% solutions** • Infusion is titrated to goal Na^+ set by primary team (typically higher values ~145-155 mEq/L) • Avoid rapid fluctuations in serum sodium (10-12 mEq/L over 24 hours) • Caution if severe hypernatremia >160 mEq/L • Monitor serum sodium q4-6 hours ❖ Abrupt changes can cause iatrogenic osmotic demyelination syndrome
23.4% sodium chloride **Administered by Neuro Critical Care provider only**	Premixed vial from pharmacy	120 mEq (30 mL) via SLOW IV push (over 10 minutes) **High concentration**	Onset: Rapid **Duration:** 1.5 – 6 hours	• **Central line administration ONLY – NO EXCEPTIONS** • Closely monitor serum sodium (q 6h)
mannitol (Osmitrol) **Use in-line filter during administration**	20 % solution (500 mL bag) = 100 grams 25% solution (50 mL vial) = 12.5 grams	0.25-2 gram/kg over 15-30 minutes Repeat q 6 hours PRN or until ICP <20	10-15 minutes Reduces ICP in 15-30 minutes **Duration:** 1.5-6 hours	• Use with caution in severe renal impairment • Inspect for crystals prior to administration: if present dissolve by warming solution under warm water • Increased UOP expected, monitor electrolytes • Do not administer in same line with blood products • Goal – serum osmolality 300-320 mOsm/kg

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IV Antiepileptics

Name	Concentrations	Dose	Onset/Duration of Action	Comments
FOSphenytoin (Cerebyx)	Variable dosing- mixed in pharmacy **D5W**	**Loading Dose**: 20-25 mg/PE per kg x1 **Maintenance Dose:** 4-6 mg PE/kg/day in divided doses (Q8- 12H) **Max Infusion Rate**: 150mg PE/min	Onset: 15 minutes Duration: dose-dependent	• Monitor for hypotension and cardiac arrhythmias during and after IV infusion. Patient should be on telemetry. • Monitor for purple glove syndrome and extravasation • Many drug interactions • Goal total phenytoin level 10-20 mcg/mL (free level 1-2 mcg/mL)
lacosamide (Vimpat)	Variable dosing mixed in pharmacy **NS**	**Loading Dose**: up to 10 mg/kg x 1 (typically max at 600 mg) **Maintenance Dose**: 200-400 mg/day in divided doses (q 12h) infused over 30 minutes	Onset: 1-4 hours Duration: 13 hours	• Renal dose adjustment required • Removed by dialysis; Supplemental dose recommended after HD • Contraindicated in patients with severe hepatic impairment • Caution in patients with cardiac conduction issues: Prolongs PR interval, may cause 1st and 2nd degree heart blocks and bradycardia • Cardiac monitoring recommended with higher doses • IV:PO conversion is 1:1 • Must waste as a narcotic
lev**ETIRA**cetam (Keppra)	**Doses ≤ 2000 mg IVP over 5 minutes** 500 mg/ 5 mL (100 mg/mL) Doses > 2000 mg will be IVPB	**Loading Dose:** 1000-4000 mg x 1 **Maintenance Dose:** 1000-3000 mg/day in divided doses Q 12 hours	Duration: 6-8 hours	• Draw up doses in 10 mL increments for IVP administration • Renal dose adjustment required • Removed by dialysis; supplemental dose recommended after HD • IV:PO conversion is 1:1 • Side Effects: Behavioral changes & somnolence
valproate sodium (Depacon)	Variable dosing mixed in pharmacy **D5W**	**Loading Dose:** 20-60 mg/kg x1 **Maintenance Dose:** up to 60 mg/kg/day in divided doses (Q6-12)	Onset: 15 minutes Duration: 9-19 hours	• Avoid use in patients with hepatic impairment • Adverse reactions include hepatotoxicity, pancreatitis, hyperammonemia, and thrombocytopenia • Goal serum valproic acid level for seizures 50-100 mcg/mL • Pregnancy category X

Antipsychotics

Name	Concentrations	Dose	Onset of Action	Duration of Action	Comments
haloperidol (Haldol)	n/a	2.5-5 mg PO, IM, or IV (may repeat Q 30-60 minutes) **Patient Dependent Max**: 10-20 mg	3-20 minutes	varies: Few minutes to 50 hours	• Can cause QT prolongation • EPS side effects (rare with IV formulation) • For acute, non-redirectable psychotic agitation • **IM** with lorazepam (compatible) • **Not compatible with diphenhydramine**
OLANZapine (Zyprexa, Zyprexa Zydis)	n/a	5-20 mg PO or IM **Initial:** 2.5-5 mg May repeat in 30 min if not effective. **Max Total Dose**: 30 mg/day	**PO:** 6 hours (peak) **IM:** 30 minutes	24-54 hours	• Can cause QT prolongation • Hypotension associated with IM formulation • **Avoid administration within 2 hours of benzodiazepines (IV or IM); may cause excessive sedation and cardiorespiratory depression**
ziprasidone (Geodon)	n/a	20-40 mg PO Q 12 hours 5-10 mg IM Q 2-4 hours **Max:** 40 mg/day	**PO:** 4 hours (peak) **IM:** <60 minutes	**PO:** 7 hours **IM:** 2.5 hours	• Can cause QT prolongation

Adult Drug Titration Reference

ATTENTION: This table is a guideline only and any titrations listed below should not supersede provider orders in Epic.

If you need to titrate differently for an individual patient, you must obtain a provider order to change the titration dose/rate.

Drug	Starting Dose	Up Titration Dose	Up Titration Time	Suggested Maximum Dose	Down Titration Dose	Down Titration Time
angiotensin II (Giapreza)	10 ng/kg/min	5-15 ng/kg/min	5 minutes	40 - 80 ng/kg/min	5-15 ng/kg/min	5-15 minutes
CISatracurium (Nimbex)	1-3 mcg/kg/min	0.5 mcg/kg/min	15 minutes	10 mcg/kg/min	0.25 mcg/kg/min	30 minutes
clevidipine (Cleviprex)	1-2 mg/hr	Double the dose	2 minutes	16 mg/hr (*32mg/hr)	1-2 mg/hr	5 minutes
dexmede**TOMID**ine (Precedex)	0.2 - 0.5 mcg/kg/hr	0.1 mcg/kg**/hr**	30 minutes	1.4 mcg/kg/hr	0.1 mcg/kg**/hr**	15 minutes
dil**TIAZ**em (Cardizem)	2.5 - 5 mg/hr	2.5 mg/hr	15 minutes	15 mg/hr	1 mg/hr	30 minutes
DOBUTamine (DOButrex)	2.5 - 10 mcg/kg/min	2.5 mcg/kg/min	15 minutes	20 mcg/kg/min	2.5 mcg/kg/min	15 minutes
DOPamine	2.5-10 mcg/kg/min	2.5 - 5 mcg/kg/min	5 minutes	20 mcg/kg/min	1 mcg/kg/min	30 minutes
EPINEPHrine	0.02 - 0.2 mcg/kg/min	0.02 mcg/kg/min	5 minutes	2 mcg/kg/min	0.01 mcg/kg/min	15 minutes
esmolol	50 mcg/kg/min	25 mcg/kg/min	5 minutes	300 mcg/kg/min	25 mcg/kg/min	15 minutes
fenta**NYL**	25 - 100 mcg/hr	12.5 mcg/hr	30 minutes	500 mcg/hr	12.5 mcg/hr	1 hour
isoproterenol (Isuprel)	2 mcg/min	0.5 - 1 mcg/min	5 minutes	10 mcg/min	1 mcg/min	30 minutes
ketamine (Ketalar)	2.5 mcg/kg/min	2.5 mcg/kg/min	15 minutes	20 mcg/kg/min	2.5 mcg/kg/min	15 minutes
labetalol	0.5 - 2 mg/min	0.1 mg/min	15 minutes	3 mg/min	0.1 mg/min	30 minutes
LIDOcaine	1 mg/min	0.5 mg/min	10 minutes	4 mg/min	0.25 mg/min	30 minutes
LORazepam (Ativan)	0.5 - 2 mg/hr	0.5 mg/hr	30 minutes	5 mg/hr	0.25 mg/hr	30 minutes
midazolam (Versed)	0.5 - 2 mg/hr	0.5 mg/hr	30 minutes	5 mg/hr	0.5 mg/hr	30 minutes
morphine	0.5 - 2 mg/hr	0.5 mg/hr	30 minutes	10 mg/hr	0.25 mg/hr	30 minutes
ni**CAR**dipine (Cardene)	2.5 - 5 mg/hr	2.5 mg/hr	5 - 15 minutes	15 mg/hr	1 mg/hr	10 minutes
nitro**GLY**cerin (Tridil)	5 - 20 mcg/min	5 - 20 mcg/min	3 - 5 minutes	400 mcg/min	5 - 20 mcg/min	15 minutes
nitro**PRUSS**ide (Nipride)	0.3 mcg/kg/min	0.2 mcg/kg/min	3 - 5 minutes	10 mcg/kg/min	0.1 mcg/kg/min	15 minutes
NORepinephrine (Levophed)	0.02 - 0.2 mcg/kg/min	0.02 mcg/kg/min	5 minutes	3 mcg/kg/min	0.02 mcg/kg/min	15 minutes
PHENYLephrine (NeoSynephrine)	0.5 mcg/kg/min	0.5 mcg/kg/min	5 minutes	5 mcg/kg/min	0.1 mcg/kg/min	15 minutes
propofol	5-10 mcg/kg/min	5 mcg/kg/min	5 minutes	50 mcg/kg/min	2.5 mcg/kg/min	15 minutes
REMIfentanil (Ultiva)	0.05 mcg/kg/min	0.025 mcg/kg/min	5 minutes	0.2 mcg/kg/min	0.025 mcg/kg/min	5 minutes
VECuronium (Norcuron)	0.5-1 mcg/kg/min	0.5 mcg/kg/min	15 minutes	2 mcg/kg/min	0.5 mcg/kg/min	15 minutes

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Notes: